USER INTERFACE DESIGN USING FLUTTER

(Common to CSE, IT, CSE(AI&ML) and AI&DS)
III Year – I Semester

Practical : 2	Internal Marks : 30
Credits : 1	External Marks : 70

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Course Objectives

- To implement Flutter widgets and layouts, including responsive UI design, navigation, and customized themed components.
- To integrate animations and data fetching to create dynamic and interactive Flutter applications.

Course Outcomes

Upon successful completion of the course, the students will be able to

- design responsive Flutter UIs using core widgets, layouts, and media queries.
- implement navigation and manage state with Navigator, named routes, and Provider.
- create custom widgets and styled forms with themes and validation.
- integrate animations and fetch data from REST APIs for dynamic UI updates.

List of Experiments:

1. a) Install Flutter and Dart SDK.
   b) Write a simple Dart program to understand the language basics.
2. a) Explore various Flutter widgets (Text, Image, Container, etc.).
   b) Implement different layout structures using Row, Column, and Stack widgets.
3. a) Design a responsive UI that adapts to different screen sizes.
   b) Implement media queries and breakpoints for responsiveness.
4. a) Set up navigation between different screens using Navigator.
   b) Implement navigation with named routes.
5. a) Learn about stateful and stateless widgets.
   b) Implement state management using set State and Provider.
6. a) Create custom widgets for specific UI elements.
   b) Apply styling using themes and custom styles.
7. a) Design a form with various input fields.
   b) Implement form validation and error handling.
8. a) Add animations to UI elements using Flutter's animation framework.
   b) Experiment with different types of animations (fade, slide, etc.).
9. a) Fetch data from a REST API.
   b) Display the fetched data in a meaningful way in the UI.
10. a) Write unit tests for UI components.
    b) Use Flutter's debugging tools to identify and fix issues.

Reference Books:

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.
2. Rap Payne, Beginning App Development with Flutter: Create Cross-Platform Mobile Apps, 1st Edition, Apress.
3. Richard Rose, Flutter & Dart Cookbook, Developing Full stack Applications for the Cloud, Oreilly.